

PBL2 L3.R

Harshal

2026-08-15

```
# Lab 3: Control Flow for Data Cleaning
getwd()
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```
## [1] "C:/SEM VII/R/PBL 2"
```

```
data <- read.csv("processed.cleveland.data", header = F, na.strings = "?")
cat(readLines("heart-disease.names"), sep = "\n")
```

Publication Request:

[illegible]

```
##      This file describes the contents of the heart-disease directory.
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##

```
##      This directory contains 4 databases concerning heart disease diagnosis.
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```
## All attributes are numeric-valued. The data was collected from the
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```
##      four following locations:
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##

```
##      1. Cleveland Clinic Foundation (cleveland.data)
```

```
##      2. Hungarian Institute of Cardiology, Budapest (hungarian.data)
```

```
##      3. V.A. Medical Center, Long Beach, CA (long-beach-va.data)
```

```
##      4. University Hospital, Zurich, Switzerland (switzerland.data)
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##

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## Each database has the same instance format. While the databases have 76
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## raw attributes, only 14 of them are actually used. Thus I've taken the
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## liberty of making 2 copies of each database: one with all the attributes
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and 1 with the 14 attributes actually used in past experiments.

##

```
## The authors of the databases have requested:
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##

```
##      ...that any publications resulting from the use of the data include the
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```
##      names of the principal investigator responsible for the data collection
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```
##          at each institution.  They would be:
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##

1. Hungarian Institute of Cardiology. Budapest: Andras Janosi, M.D.

2. University Hospital, Zurich, Switzerland: William Steinbrunn, M.D.

3. University Hospital, Basel, Switzerland: Matthias Pfisterer, M.D.

4. V.A. Medical Center, Long Beach and Cleveland Clinic Foundation:

Robert Detrano, M.D., Ph.D.

##

Thanks in advance for abiding by this request.

##


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##      database is the only one that has been used by ML researchers to
##      this date.  The "goal" field refers to the presence of heart disease
##      in the patient.  It is integer valued from 0 (no presence) to 4.
##      Experiments with the Cleveland database have concentrated on simply
##      attempting to distinguish presence (values 1,2,3,4) from absence (value
##      0).
##
##      The names and social security numbers of the patients were recently
##      removed from the database, replaced with dummy values.
##
##      One file has been "processed", that one containing the Cleveland
##      database.  All four unprocessed files also exist in this directory.
##
## 5. Number of Instances:
##      Database:      # of instances:
##      Cleveland: 303
##      Hungarian: 294
##      Switzerland: 123
##      Long Beach VA: 200
##
## 6. Number of Attributes: 76 (including the predicted attribute)
##
## 7. Attribute Information:
##      -- Only 14 used
##      -- 1. #3  (age)
##      -- 2. #4  (sex)
##      -- 3. #9  (cp)
##      -- 4. #10 (trestbps)
##      -- 5. #12 (chol)
##      -- 6. #16 (fbs)
##      -- 7. #19 (restecg)
##      -- 8. #32 (thalach)
##      -- 9. #38 (exang)
##      -- 10. #40 (oldpeak)
##      -- 11. #41 (slope)
##      -- 12. #44 (ca)
##      -- 13. #51 (thal)
##      -- 14. #58 (num)          (the predicted attribute)
##
##      -- Complete attribute documentation:
##      1 id: patient identification number
##      2 ccf: social security number (I replaced this with a dummy value of 0)
##      3 age: age in years
##      4 sex: sex (1 = male; 0 = female)
##      5 painloc: chest pain location (1 = substernal; 0 = otherwise)
##      6 painexer (1 = provoked by exertion; 0 = otherwise)
##      7 relrest (1 = relieved after rest; 0 = otherwise)
##      8 pncaden (sum of 5, 6, and 7)
##      9 cp: chest pain type
##      -- Value 1: typical angina
##      -- Value 2: atypical angina
##      -- Value 3: non-anginal pain
##      -- Value 4: asymptomatic
##      10 trestbps: resting blood pressure (in mm Hg on admission to the

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##      hospital)
## 11 htn
## 12 chol: serum cholestoral in mg/dl
## 13 smoke: I believe this is 1 = yes; 0 = no (is or is not a smoker)
## 14 cigs (cigarettes per day)
## 15 years (number of years as a smoker)
## 16 fbs: (fasting blood sugar > 120 mg/dl) (1 = true; 0 = false)
## 17 dm (1 = history of diabetes; 0 = no such history)
## 18 famhist: family history of coronary artery disease (1 = yes; 0 = no)
## 19 restecg: resting electrocardiographic results
##      -- Value 0: normal
##      -- Value 1: having ST-T wave abnormality (T wave inversions and/or ST
##                  elevation or depression of > 0.05 mV)
##      -- Value 2: showing probable or definite left ventricular hypertrophy
##                  by Estes' criteria
## 20 ekgmo (month of exercise ECG reading)
## 21 ekgday(day of exercise ECG reading)
## 22 ekgyr (year of exercise ECG reading)
## 23 dig (digitalis used during exercise ECG: 1 = yes; 0 = no)
## 24 prop (Beta blocker used during exercise ECG: 1 = yes; 0 = no)
## 25 nitr (nitrates used during exercise ECG: 1 = yes; 0 = no)
## 26 pro (calcium channel blocker used during exercise ECG: 1 = yes; 0 = no)
## 27 diuretic (diuretic used used during exercise ECG: 1 = yes; 0 = no)
## 28 proto: exercise protocol
##      1 = Bruce
##      2 = Kottus
##      3 = McHenry
##      4 = fast Balke
##      5 = Balke
##      6 = Noughton
##      7 = bike 150 kpa min/min (Not sure if "kpa min/min" is what was
##                              written!)
##      8 = bike 125 kpa min/min
##      9 = bike 100 kpa min/min
##     10 = bike 75 kpa min/min
##     11 = bike 50 kpa min/min
##     12 = arm ergometer
## 29 thaldur: duration of exercise test in minutes
## 30 thaltime: time when ST measure depression was noted
## 31 met: mets achieved
## 32 thalach: maximum heart rate achieved
## 33 thalrest: resting heart rate
## 34 tpeakbps: peak exercise blood pressure (first of 2 parts)
## 35 tpeakbpd: peak exercise blood pressure (second of 2 parts)
## 36 dummy
## 37 trestbpd: resting blood pressure
## 38 exang: exercise induced angina (1 = yes; 0 = no)
## 39 xhypo: (1 = yes; 0 = no)
## 40 oldpeak = ST depression induced by exercise relative to rest
## 41 slope: the slope of the peak exercise ST segment
##      -- Value 1: upsloping
##      -- Value 2: flat
##      -- Value 3: downsloping
## 42 rldv5: height at rest

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##      43 rldv5e: height at peak exercise
##      44 ca: number of major vessels (0-3) colored by flourosopy
##      45 restckm: irrelevant
##      46 exerckm: irrelevant
##      47 restef: rest raidonuclid (sp?) ejection fraction
##      48 restwm: rest wall (sp?) motion abnormality
##          0 = none
##          1 = mild or moderate
##          2 = moderate or severe
##          3 = akinesis or dyskmem (sp?)
##      49 exeref: exercise radinalid (sp?) ejection fraction
##      50 exerwm: exercise wall (sp?) motion
##      51 thal: 3 = normal; 6 = fixed defect; 7 = reversable defect
##      52 thalsev: not used
##      53 thalpul: not used
##      54 earlobe: not used
##      55 cmo: month of cardiac cath (sp?) (perhaps "call")
##      56 cday: day of cardiac cath (sp?)
##      57 cyr: year of cardiac cath (sp?)
##      58 num: diagnosis of heart disease (angiographic disease status)
##          -- Value 0: < 50% diameter narrowing
##          -- Value 1: > 50% diameter narrowing
##          (in any major vessel: attributes 59 through 68 are vessels)
##      59 lmt
##      60 ladprox
##      61 laddist
##      62 diag
##      63 cxmain
##      64 ramus
##      65 om1
##      66 om2
##      67 rcaprox
##      68 rcadist
##      69 lvx1: not used
##      70 lvx2: not used
##      71 lvx3: not used
##      72 lvx4: not used
##      73 lvf: not used
##      74 cathef: not used
##      75 junk: not used
##      76 name: last name of patient
##      (I replaced this with the dummy string "name")
##
## 9. Missing Attribute Values: Several. Distinguished with value -9.0.
##
## 10. Class Distribution:
##      Database:      0    1    2    3    4 Total
##      Cleveland: 164  55  36  35  13  303
##      Hungarian: 188  37  26  28  15  294
##      Switzerland:  8  48  32  30   5  123
##      Long Beach VA: 51  56  41  42  10  200

```

data

##	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	V13	V14
## 1	63	1	1	145	233	1	2	150	0	2.3	3	0	6	0
## 2	67	1	4	160	286	0	2	108	1	1.5	2	3	3	2
## 3	67	1	4	120	229	0	2	129	1	2.6	2	2	7	1
## 4	37	1	3	130	250	0	0	187	0	3.5	3	0	3	0
## 5	41	0	2	130	204	0	2	172	0	1.4	1	0	3	0
## 6	56	1	2	120	236	0	0	178	0	0.8	1	0	3	0
## 7	62	0	4	140	268	0	2	160	0	3.6	3	2	3	3
## 8	57	0	4	120	354	0	0	163	1	0.6	1	0	3	0
## 9	63	1	4	130	254	0	2	147	0	1.4	2	1	7	2
## 10	53	1	4	140	203	1	2	155	1	3.1	3	0	7	1
## 11	57	1	4	140	192	0	0	148	0	0.4	2	0	6	0
## 12	56	0	2	140	294	0	2	153	0	1.3	2	0	3	0
## 13	56	1	3	130	256	1	2	142	1	0.6	2	1	6	2
## 14	44	1	2	120	263	0	0	173	0	0.0	1	0	7	0
## 15	52	1	3	172	199	1	0	162	0	0.5	1	0	7	0
## 16	57	1	3	150	168	0	0	174	0	1.6	1	0	3	0
## 17	48	1	2	110	229	0	0	168	0	1.0	3	0	7	1
## 18	54	1	4	140	239	0	0	160	0	1.2	1	0	3	0
## 19	48	0	3	130	275	0	0	139	0	0.2	1	0	3	0
## 20	49	1	2	130	266	0	0	171	0	0.6	1	0	3	0
## 21	64	1	1	110	211	0	2	144	1	1.8	2	0	3	0
## 22	58	0	1	150	283	1	2	162	0	1.0	1	0	3	0
## 23	58	1	2	120	284	0	2	160	0	1.8	2	0	3	1
## 24	58	1	3	132	224	0	2	173	0	3.2	1	2	7	3
## 25	60	1	4	130	206	0	2	132	1	2.4	2	2	7	4
## 26	50	0	3	120	219	0	0	158	0	1.6	2	0	3	0
## 27	58	0	3	120	340	0	0	172	0	0.0	1	0	3	0
## 28	66	0	1	150	226	0	0	114	0	2.6	3	0	3	0
## 29	43	1	4	150	247	0	0	171	0	1.5	1	0	3	0
## 30	40	1	4	110	167	0	2	114	1	2.0	2	0	7	3
## 31	69	0	1	140	239	0	0	151	0	1.8	1	2	3	0
## 32	60	1	4	117	230	1	0	160	1	1.4	1	2	7	2
## 33	64	1	3	140	335	0	0	158	0	0.0	1	0	3	1
## 34	59	1	4	135	234	0	0	161	0	0.5	2	0	7	0
## 35	44	1	3	130	233	0	0	179	1	0.4	1	0	3	0
## 36	42	1	4	140	226	0	0	178	0	0.0	1	0	3	0
## 37	43	1	4	120	177	0	2	120	1	2.5	2	0	7	3
## 38	57	1	4	150	276	0	2	112	1	0.6	2	1	6	1
## 39	55	1	4	132	353	0	0	132	1	1.2	2	1	7	3
## 40	61	1	3	150	243	1	0	137	1	1.0	2	0	3	0
## 41	65	0	4	150	225	0	2	114	0	1.0	2	3	7	4
## 42	40	1	1	140	199	0	0	178	1	1.4	1	0	7	0
## 43	71	0	2	160	302	0	0	162	0	0.4	1	2	3	0
## 44	59	1	3	150	212	1	0	157	0	1.6	1	0	3	0
## 45	61	0	4	130	330	0	2	169	0	0.0	1	0	3	1
## 46	58	1	3	112	230	0	2	165	0	2.5	2	1	7	4
## 47	51	1	3	110	175	0	0	123	0	0.6	1	0	3	0
## 48	50	1	4	150	243	0	2	128	0	2.6	2	0	7	4
## 49	65	0	3	140	417	1	2	157	0	0.8	1	1	3	0
## 50	53	1	3	130	197	1	2	152	0	1.2	3	0	3	0
## 51	41	0	2	105	198	0	0	168	0	0.0	1	1	3	0

##	52	65	1	4	120	177	0	0	140	0	0.4	1	0	7	0
##	53	44	1	4	112	290	0	2	153	0	0.0	1	1	3	2
##	54	44	1	2	130	219	0	2	188	0	0.0	1	0	3	0
##	55	60	1	4	130	253	0	0	144	1	1.4	1	1	7	1
##	56	54	1	4	124	266	0	2	109	1	2.2	2	1	7	1
##	57	50	1	3	140	233	0	0	163	0	0.6	2	1	7	1
##	58	41	1	4	110	172	0	2	158	0	0.0	1	0	7	1
##	59	54	1	3	125	273	0	2	152	0	0.5	3	1	3	0
##	60	51	1	1	125	213	0	2	125	1	1.4	1	1	3	0
##	61	51	0	4	130	305	0	0	142	1	1.2	2	0	7	2
##	62	46	0	3	142	177	0	2	160	1	1.4	3	0	3	0
##	63	58	1	4	128	216	0	2	131	1	2.2	2	3	7	1
##	64	54	0	3	135	304	1	0	170	0	0.0	1	0	3	0
##	65	54	1	4	120	188	0	0	113	0	1.4	2	1	7	2
##	66	60	1	4	145	282	0	2	142	1	2.8	2	2	7	2
##	67	60	1	3	140	185	0	2	155	0	3.0	2	0	3	1
##	68	54	1	3	150	232	0	2	165	0	1.6	1	0	7	0
##	69	59	1	4	170	326	0	2	140	1	3.4	3	0	7	2
##	70	46	1	3	150	231	0	0	147	0	3.6	2	0	3	1
##	71	65	0	3	155	269	0	0	148	0	0.8	1	0	3	0
##	72	67	1	4	125	254	1	0	163	0	0.2	2	2	7	3
##	73	62	1	4	120	267	0	0	99	1	1.8	2	2	7	1
##	74	65	1	4	110	248	0	2	158	0	0.6	1	2	6	1
##	75	44	1	4	110	197	0	2	177	0	0.0	1	1	3	1
##	76	65	0	3	160	360	0	2	151	0	0.8	1	0	3	0
##	77	60	1	4	125	258	0	2	141	1	2.8	2	1	7	1
##	78	51	0	3	140	308	0	2	142	0	1.5	1	1	3	0
##	79	48	1	2	130	245	0	2	180	0	0.2	2	0	3	0
##	80	58	1	4	150	270	0	2	111	1	0.8	1	0	7	3
##	81	45	1	4	104	208	0	2	148	1	3.0	2	0	3	0
##	82	53	0	4	130	264	0	2	143	0	0.4	2	0	3	0
##	83	39	1	3	140	321	0	2	182	0	0.0	1	0	3	0
##	84	68	1	3	180	274	1	2	150	1	1.6	2	0	7	3
##	85	52	1	2	120	325	0	0	172	0	0.2	1	0	3	0
##	86	44	1	3	140	235	0	2	180	0	0.0	1	0	3	0
##	87	47	1	3	138	257	0	2	156	0	0.0	1	0	3	0
##	88	53	0	3	128	216	0	2	115	0	0.0	1	0	NA	0
##	89	53	0	4	138	234	0	2	160	0	0.0	1	0	3	0
##	90	51	0	3	130	256	0	2	149	0	0.5	1	0	3	0
##	91	66	1	4	120	302	0	2	151	0	0.4	2	0	3	0
##	92	62	0	4	160	164	0	2	145	0	6.2	3	3	7	3
##	93	62	1	3	130	231	0	0	146	0	1.8	2	3	7	0
##	94	44	0	3	108	141	0	0	175	0	0.6	2	0	3	0
##	95	63	0	3	135	252	0	2	172	0	0.0	1	0	3	0
##	96	52	1	4	128	255	0	0	161	1	0.0	1	1	7	1
##	97	59	1	4	110	239	0	2	142	1	1.2	2	1	7	2
##	98	60	0	4	150	258	0	2	157	0	2.6	2	2	7	3
##	99	52	1	2	134	201	0	0	158	0	0.8	1	1	3	0
##	100	48	1	4	122	222	0	2	186	0	0.0	1	0	3	0
##	101	45	1	4	115	260	0	2	185	0	0.0	1	0	3	0
##	102	34	1	1	118	182	0	2	174	0	0.0	1	0	3	0
##	103	57	0	4	128	303	0	2	159	0	0.0	1	1	3	0
##	104	71	0	3	110	265	1	2	130	0	0.0	1	1	3	0
##	105	49	1	3	120	188	0	0	139	0	2.0	2	3	7	3

##	106	54	1	2	108	309	0	0	156	0	0.0	1	0	7	0
##	107	59	1	4	140	177	0	0	162	1	0.0	1	1	7	2
##	108	57	1	3	128	229	0	2	150	0	0.4	2	1	7	1
##	109	61	1	4	120	260	0	0	140	1	3.6	2	1	7	2
##	110	39	1	4	118	219	0	0	140	0	1.2	2	0	7	3
##	111	61	0	4	145	307	0	2	146	1	1.0	2	0	7	1
##	112	56	1	4	125	249	1	2	144	1	1.2	2	1	3	1
##	113	52	1	1	118	186	0	2	190	0	0.0	2	0	6	0
##	114	43	0	4	132	341	1	2	136	1	3.0	2	0	7	2
##	115	62	0	3	130	263	0	0	97	0	1.2	2	1	7	2
##	116	41	1	2	135	203	0	0	132	0	0.0	2	0	6	0
##	117	58	1	3	140	211	1	2	165	0	0.0	1	0	3	0
##	118	35	0	4	138	183	0	0	182	0	1.4	1	0	3	0
##	119	63	1	4	130	330	1	2	132	1	1.8	1	3	7	3
##	120	65	1	4	135	254	0	2	127	0	2.8	2	1	7	2
##	121	48	1	4	130	256	1	2	150	1	0.0	1	2	7	3
##	122	63	0	4	150	407	0	2	154	0	4.0	2	3	7	4
##	123	51	1	3	100	222	0	0	143	1	1.2	2	0	3	0
##	124	55	1	4	140	217	0	0	111	1	5.6	3	0	7	3
##	125	65	1	1	138	282	1	2	174	0	1.4	2	1	3	1
##	126	45	0	2	130	234	0	2	175	0	0.6	2	0	3	0
##	127	56	0	4	200	288	1	2	133	1	4.0	3	2	7	3
##	128	54	1	4	110	239	0	0	126	1	2.8	2	1	7	3
##	129	44	1	2	120	220	0	0	170	0	0.0	1	0	3	0
##	130	62	0	4	124	209	0	0	163	0	0.0	1	0	3	0
##	131	54	1	3	120	258	0	2	147	0	0.4	2	0	7	0
##	132	51	1	3	94	227	0	0	154	1	0.0	1	1	7	0
##	133	29	1	2	130	204	0	2	202	0	0.0	1	0	3	0
##	134	51	1	4	140	261	0	2	186	1	0.0	1	0	3	0
##	135	43	0	3	122	213	0	0	165	0	0.2	2	0	3	0
##	136	55	0	2	135	250	0	2	161	0	1.4	2	0	3	0
##	137	70	1	4	145	174	0	0	125	1	2.6	3	0	7	4
##	138	62	1	2	120	281	0	2	103	0	1.4	2	1	7	3
##	139	35	1	4	120	198	0	0	130	1	1.6	2	0	7	1
##	140	51	1	3	125	245	1	2	166	0	2.4	2	0	3	0
##	141	59	1	2	140	221	0	0	164	1	0.0	1	0	3	0
##	142	59	1	1	170	288	0	2	159	0	0.2	2	0	7	1
##	143	52	1	2	128	205	1	0	184	0	0.0	1	0	3	0
##	144	64	1	3	125	309	0	0	131	1	1.8	2	0	7	1
##	145	58	1	3	105	240	0	2	154	1	0.6	2	0	7	0
##	146	47	1	3	108	243	0	0	152	0	0.0	1	0	3	1
##	147	57	1	4	165	289	1	2	124	0	1.0	2	3	7	4
##	148	41	1	3	112	250	0	0	179	0	0.0	1	0	3	0
##	149	45	1	2	128	308	0	2	170	0	0.0	1	0	3	0
##	150	60	0	3	102	318	0	0	160	0	0.0	1	1	3	0
##	151	52	1	1	152	298	1	0	178	0	1.2	2	0	7	0
##	152	42	0	4	102	265	0	2	122	0	0.6	2	0	3	0
##	153	67	0	3	115	564	0	2	160	0	1.6	2	0	7	0
##	154	55	1	4	160	289	0	2	145	1	0.8	2	1	7	4
##	155	64	1	4	120	246	0	2	96	1	2.2	3	1	3	3
##	156	70	1	4	130	322	0	2	109	0	2.4	2	3	3	1
##	157	51	1	4	140	299	0	0	173	1	1.6	1	0	7	1
##	158	58	1	4	125	300	0	2	171	0	0.0	1	2	7	1
##	159	60	1	4	140	293	0	2	170	0	1.2	2	2	7	2

##	160	68	1	3	118	277	0	0	151	0	1.0	1	1	7	0
##	161	46	1	2	101	197	1	0	156	0	0.0	1	0	7	0
##	162	77	1	4	125	304	0	2	162	1	0.0	1	3	3	4
##	163	54	0	3	110	214	0	0	158	0	1.6	2	0	3	0
##	164	58	0	4	100	248	0	2	122	0	1.0	2	0	3	0
##	165	48	1	3	124	255	1	0	175	0	0.0	1	2	3	0
##	166	57	1	4	132	207	0	0	168	1	0.0	1	0	7	0
##	167	52	1	3	138	223	0	0	169	0	0.0	1	NA	3	0
##	168	54	0	2	132	288	1	2	159	1	0.0	1	1	3	0
##	169	35	1	4	126	282	0	2	156	1	0.0	1	0	7	1
##	170	45	0	2	112	160	0	0	138	0	0.0	2	0	3	0
##	171	70	1	3	160	269	0	0	112	1	2.9	2	1	7	3
##	172	53	1	4	142	226	0	2	111	1	0.0	1	0	7	0
##	173	59	0	4	174	249	0	0	143	1	0.0	2	0	3	1
##	174	62	0	4	140	394	0	2	157	0	1.2	2	0	3	0
##	175	64	1	4	145	212	0	2	132	0	2.0	2	2	6	4
##	176	57	1	4	152	274	0	0	88	1	1.2	2	1	7	1
##	177	52	1	4	108	233	1	0	147	0	0.1	1	3	7	0
##	178	56	1	4	132	184	0	2	105	1	2.1	2	1	6	1
##	179	43	1	3	130	315	0	0	162	0	1.9	1	1	3	0
##	180	53	1	3	130	246	1	2	173	0	0.0	1	3	3	0
##	181	48	1	4	124	274	0	2	166	0	0.5	2	0	7	3
##	182	56	0	4	134	409	0	2	150	1	1.9	2	2	7	2
##	183	42	1	1	148	244	0	2	178	0	0.8	1	2	3	0
##	184	59	1	1	178	270	0	2	145	0	4.2	3	0	7	0
##	185	60	0	4	158	305	0	2	161	0	0.0	1	0	3	1
##	186	63	0	2	140	195	0	0	179	0	0.0	1	2	3	0
##	187	42	1	3	120	240	1	0	194	0	0.8	3	0	7	0
##	188	66	1	2	160	246	0	0	120	1	0.0	2	3	6	2
##	189	54	1	2	192	283	0	2	195	0	0.0	1	1	7	1
##	190	69	1	3	140	254	0	2	146	0	2.0	2	3	7	2
##	191	50	1	3	129	196	0	0	163	0	0.0	1	0	3	0
##	192	51	1	4	140	298	0	0	122	1	4.2	2	3	7	3
##	193	43	1	4	132	247	1	2	143	1	0.1	2	NA	7	1
##	194	62	0	4	138	294	1	0	106	0	1.9	2	3	3	2
##	195	68	0	3	120	211	0	2	115	0	1.5	2	0	3	0
##	196	67	1	4	100	299	0	2	125	1	0.9	2	2	3	3
##	197	69	1	1	160	234	1	2	131	0	0.1	2	1	3	0
##	198	45	0	4	138	236	0	2	152	1	0.2	2	0	3	0
##	199	50	0	2	120	244	0	0	162	0	1.1	1	0	3	0
##	200	59	1	1	160	273	0	2	125	0	0.0	1	0	3	1
##	201	50	0	4	110	254	0	2	159	0	0.0	1	0	3	0
##	202	64	0	4	180	325	0	0	154	1	0.0	1	0	3	0
##	203	57	1	3	150	126	1	0	173	0	0.2	1	1	7	0
##	204	64	0	3	140	313	0	0	133	0	0.2	1	0	7	0
##	205	43	1	4	110	211	0	0	161	0	0.0	1	0	7	0
##	206	45	1	4	142	309	0	2	147	1	0.0	2	3	7	3
##	207	58	1	4	128	259	0	2	130	1	3.0	2	2	7	3
##	208	50	1	4	144	200	0	2	126	1	0.9	2	0	7	3
##	209	55	1	2	130	262	0	0	155	0	0.0	1	0	3	0
##	210	62	0	4	150	244	0	0	154	1	1.4	2	0	3	1
##	211	37	0	3	120	215	0	0	170	0	0.0	1	0	3	0
##	212	38	1	1	120	231	0	0	182	1	3.8	2	0	7	4
##	213	41	1	3	130	214	0	2	168	0	2.0	2	0	3	0

##	214	66	0	4	178	228	1	0	165	1	1.0	2	2	7	3
##	215	52	1	4	112	230	0	0	160	0	0.0	1	1	3	1
##	216	56	1	1	120	193	0	2	162	0	1.9	2	0	7	0
##	217	46	0	2	105	204	0	0	172	0	0.0	1	0	3	0
##	218	46	0	4	138	243	0	2	152	1	0.0	2	0	3	0
##	219	64	0	4	130	303	0	0	122	0	2.0	2	2	3	0
##	220	59	1	4	138	271	0	2	182	0	0.0	1	0	3	0
##	221	41	0	3	112	268	0	2	172	1	0.0	1	0	3	0
##	222	54	0	3	108	267	0	2	167	0	0.0	1	0	3	0
##	223	39	0	3	94	199	0	0	179	0	0.0	1	0	3	0
##	224	53	1	4	123	282	0	0	95	1	2.0	2	2	7	3
##	225	63	0	4	108	269	0	0	169	1	1.8	2	2	3	1
##	226	34	0	2	118	210	0	0	192	0	0.7	1	0	3	0
##	227	47	1	4	112	204	0	0	143	0	0.1	1	0	3	0
##	228	67	0	3	152	277	0	0	172	0	0.0	1	1	3	0
##	229	54	1	4	110	206	0	2	108	1	0.0	2	1	3	3
##	230	66	1	4	112	212	0	2	132	1	0.1	1	1	3	2
##	231	52	0	3	136	196	0	2	169	0	0.1	2	0	3	0
##	232	55	0	4	180	327	0	1	117	1	3.4	2	0	3	2
##	233	49	1	3	118	149	0	2	126	0	0.8	1	3	3	1
##	234	74	0	2	120	269	0	2	121	1	0.2	1	1	3	0
##	235	54	0	3	160	201	0	0	163	0	0.0	1	1	3	0
##	236	54	1	4	122	286	0	2	116	1	3.2	2	2	3	3
##	237	56	1	4	130	283	1	2	103	1	1.6	3	0	7	2
##	238	46	1	4	120	249	0	2	144	0	0.8	1	0	7	1
##	239	49	0	2	134	271	0	0	162	0	0.0	2	0	3	0
##	240	42	1	2	120	295	0	0	162	0	0.0	1	0	3	0
##	241	41	1	2	110	235	0	0	153	0	0.0	1	0	3	0
##	242	41	0	2	126	306	0	0	163	0	0.0	1	0	3	0
##	243	49	0	4	130	269	0	0	163	0	0.0	1	0	3	0
##	244	61	1	1	134	234	0	0	145	0	2.6	2	2	3	2
##	245	60	0	3	120	178	1	0	96	0	0.0	1	0	3	0
##	246	67	1	4	120	237	0	0	71	0	1.0	2	0	3	2
##	247	58	1	4	100	234	0	0	156	0	0.1	1	1	7	2
##	248	47	1	4	110	275	0	2	118	1	1.0	2	1	3	1
##	249	52	1	4	125	212	0	0	168	0	1.0	1	2	7	3
##	250	62	1	2	128	208	1	2	140	0	0.0	1	0	3	0
##	251	57	1	4	110	201	0	0	126	1	1.5	2	0	6	0
##	252	58	1	4	146	218	0	0	105	0	2.0	2	1	7	1
##	253	64	1	4	128	263	0	0	105	1	0.2	2	1	7	0
##	254	51	0	3	120	295	0	2	157	0	0.6	1	0	3	0
##	255	43	1	4	115	303	0	0	181	0	1.2	2	0	3	0
##	256	42	0	3	120	209	0	0	173	0	0.0	2	0	3	0
##	257	67	0	4	106	223	0	0	142	0	0.3	1	2	3	0
##	258	76	0	3	140	197	0	1	116	0	1.1	2	0	3	0
##	259	70	1	2	156	245	0	2	143	0	0.0	1	0	3	0
##	260	57	1	2	124	261	0	0	141	0	0.3	1	0	7	1
##	261	44	0	3	118	242	0	0	149	0	0.3	2	1	3	0
##	262	58	0	2	136	319	1	2	152	0	0.0	1	2	3	3
##	263	60	0	1	150	240	0	0	171	0	0.9	1	0	3	0
##	264	44	1	3	120	226	0	0	169	0	0.0	1	0	3	0
##	265	61	1	4	138	166	0	2	125	1	3.6	2	1	3	4
##	266	42	1	4	136	315	0	0	125	1	1.8	2	0	6	2
##	267	52	1	4	128	204	1	0	156	1	1.0	2	0	NA	2

```
## 268 59 1 3 126 218 1 0 134 0 2.2 2 1 6 2
## 269 40 1 4 152 223 0 0 181 0 0.0 1 0 7 1
## 270 42 1 3 130 180 0 0 150 0 0.0 1 0 3 0
## 271 61 1 4 140 207 0 2 138 1 1.9 1 1 7 1
## 272 66 1 4 160 228 0 2 138 0 2.3 1 0 6 0
## 273 46 1 4 140 311 0 0 120 1 1.8 2 2 7 2
## 274 71 0 4 112 149 0 0 125 0 1.6 2 0 3 0
## 275 59 1 1 134 204 0 0 162 0 0.8 1 2 3 1
## 276 64 1 1 170 227 0 2 155 0 0.6 2 0 7 0
## 277 66 0 3 146 278 0 2 152 0 0.0 2 1 3 0
## 278 39 0 3 138 220 0 0 152 0 0.0 2 0 3 0
## 279 57 1 2 154 232 0 2 164 0 0.0 1 1 3 1
## 280 58 0 4 130 197 0 0 131 0 0.6 2 0 3 0
## 281 57 1 4 110 335 0 0 143 1 3.0 2 1 7 2
## 282 47 1 3 130 253 0 0 179 0 0.0 1 0 3 0
## 283 55 0 4 128 205 0 1 130 1 2.0 2 1 7 3
## 284 35 1 2 122 192 0 0 174 0 0.0 1 0 3 0
## 285 61 1 4 148 203 0 0 161 0 0.0 1 1 7 2
## 286 58 1 4 114 318 0 1 140 0 4.4 3 3 6 4
## 287 58 0 4 170 225 1 2 146 1 2.8 2 2 6 2
## 288 58 1 2 125 220 0 0 144 0 0.4 2 NA 7 0
## 289 56 1 2 130 221 0 2 163 0 0.0 1 0 7 0
## 290 56 1 2 120 240 0 0 169 0 0.0 3 0 3 0
## 291 67 1 3 152 212 0 2 150 0 0.8 2 0 7 1
## 292 55 0 2 132 342 0 0 166 0 1.2 1 0 3 0
## 293 44 1 4 120 169 0 0 144 1 2.8 3 0 6 2
## 294 63 1 4 140 187 0 2 144 1 4.0 1 2 7 2
## 295 63 0 4 124 197 0 0 136 1 0.0 2 0 3 1
## 296 41 1 2 120 157 0 0 182 0 0.0 1 0 3 0
## 297 59 1 4 164 176 1 2 90 0 1.0 2 2 6 3
## 298 57 0 4 140 241 0 0 123 1 0.2 2 0 7 1
## 299 45 1 1 110 264 0 0 132 0 1.2 2 0 7 1
## 300 68 1 4 144 193 1 0 141 0 3.4 2 2 7 2
## 301 57 1 4 130 131 0 0 115 1 1.2 2 1 7 3
## 302 57 0 2 130 236 0 2 174 0 0.0 2 1 3 1
## 303 38 1 3 138 175 0 0 173 0 0.0 1 NA 3 0
```

```
colnames(data) <- c("age", "sex", "cp", "trestbps", "chol", "fbs",
                   "restecg", "thalach", "exang", "oldpeak",
                   "slope", "ca", "thal", "num")
# above were taken from o/p of heart-disease.names
# purposeful anomalies as stated
{
  set.seed(47)

  n <- nrow(data)

  n # num of rows we got

  data$trestbps[sample(1:n, 5)] <- -150
  data$trestbps[sample(1:n, 5)] <- NA
  data$trestbps[sample(1:n, 5)] <- 340
}
```

```

# task 1 bp cleaning func

bpClean <- function(bp) {
  if(is.na(bp))
    return(NA)
  else if(bp<0)
    return(NA)
  else if(bp > 250)
    return(250)
  else
    return(bp)
}

# task 2 error handling using try catch

calcBP <- function(chol, trestbps) {
  tryCatch(
    expr = {
      if(is.na(trestbps) | is.na(chol)) stop("Input Value is NA")
      if(trestbps <= 0) stop("Denominator is <= 0")
      return(chol / trestbps)
    },
    error = function(e) {
      warning(paste("Handled Error: ", e$message))
      return(NA)
    }
  )
}

meanBP <-
  tryCatch(
    expr = mean(data$trestbps, na.rm = T),
    error = function(e) {message("Error calculating mean:", e$message); return(NA)}
  )

# task 3 loop based vs vectorized data cleaning

loop <- function(bp) {
  result <- numeric(length(bp))
  for (i in seq_along(bp))
    result[i] <- bpClean(bp[i])
  return(result)
}

vectorized <- function(bp) {
  bp[bp < 0] <- NA
  bp[bp > 250] <- 250
  return(bp)
}

if (!require(microbenchmark)) install.packages("microbenchmark")

```

```
## Loading required package: microbenchmark
```

```
library(microbenchmark)

benchmark_results <- microbenchmark(
  Loop = loop(data$trestbps),
  Vectorized = vectorized(data$trestbps),
  times = 100
)
print(benchmark_results)
```

```
## Unit: microseconds
##      expr    min      lq    mean median      uq    max neval
##      Loop 148.9 158.25 231.318 162.6 177.65 6416.9   100
## Vectorized   4.5   5.00  29.268   5.6   7.10 2144.0   100
```

```
# task 4 validation of cleaned data after above steps
```

```
cat("Negative BP:", sum(data$trestbps < 0, na.rm = TRUE), "\n")
```

```
## Negative BP: 5
```

```
cat("BP > 250:", sum(data$trestbps > 250, na.rm = TRUE), "\n")
```

```
## BP > 250: 5
```

```
data$trestbps <- vectorized(data$trestbps)

cat("Missing BP Count:", sum(is.na(data$trestbps)), "\n")
```

```
## Missing BP Count: 10
```

```
cat("Min BP:", min(data$trestbps, na.rm = TRUE), "\n")
```

```
## Min BP: 94
```

```
cat("Max BP:", max(data$trestbps, na.rm = TRUE), "\n")
```

```
## Max BP: 250
```

```
cat("Mean BP:", mean(data$trestbps, na.rm = TRUE), "\n")
```

```
## Mean BP: 133.43
```

```
cat("Median BP:", median(data$trestbps, na.rm = TRUE), "\n")
```

```
## Median BP: 130
```

```
cat("Remaining Negative BP:", sum(data$trestbps < 0, na.rm = TRUE), "\n")
```

```
## Remaining Negative BP: 0
```

```
cat("Remaining BP > 250:", sum(data$trestbps > 250, na.rm = TRUE), "\n")
```

```
## Remaining BP > 250: 0
```

```
write.csv(data, "cleaned_heart_data.csv", row.names = FALSE)
```

```
print("Vectorized cleaning is much faster than loop  
      (approximately 7.5x faster by mean of time taken)")
```

```
## [1] "Vectorized cleaning is much faster than loop \n      (approximately 7.5x faster by mean of time
```

```
print("This is because a for loop iterates line by line to perform  
      calculations for each entry in R's interpreted env.  
      Meanwhile, vectorized computation offloads this task to  
      pre-compiled hardware optimized code specifically meant  
      for this task. It skips the time overhead of interpretation by  
      directly sending the data to the underlying language.")
```

```
## [1] "This is because a for loop iterates line by line to perform \n      calculations for each entry
```